2026 UNIDROIT Essay Competition

Regulating AI-Driven Smart Contracts in Cross-Border Commerce: The Case for a UNIDROIT Model Framework

Category B | Workstream: Financial Markets and Technology

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Abstract

Artificial intelligence (AI)-driven smart contracts are rapidly transforming cross-border commercial transactions, yet the existing international private law architecture — including the UNIDROIT Principles on International Commercial Contracts (PICC) — was designed for human parties and fails to address the unique legal challenges posed by autonomous AI agents. This essay identifies critical legal gaps across five doctrinal areas — legal capacity, contract formation, liability attribution, jurisdiction, and dispute resolution — and proposes a targeted UNIDROIT Model Protocol to bridge these gaps. Drawing on comparative analysis of the EU AI Act, UNCITRAL e-Commerce instruments, and existing UNIDROIT frameworks, this essay argues that only a dedicated, harmonised international instrument can provide the legal certainty necessary for inclusive and sustainable global commerce in the age of AI.

1. Introduction

The global commercial landscape is witnessing an unprecedented transformation driven by artificial intelligence. AI agents — software systems capable of autonomous decision-making — now independently negotiate, execute, and enforce commercial contracts without human intervention at each transactional step. These *AI-driven smart contracts*, self-executing agreements encoded on distributed ledgers, are projected to constitute a significant share of international commercial transactions by 2030.

Yet the legal framework governing them remains dangerously fragmented. Traditional contract law doctrines presuppose human actors with legal personhood, intentionality, and capacity — attributes that AI systems do not unambiguously possess. UNIDROIT's centenary presents a unique opportunity: to extend its century-long mission of harmonising private law into this critical frontier. This essay argues for a dedicated UNIDROIT Model Protocol on AI-executed contracts within the Financial Markets and Technology workstream.

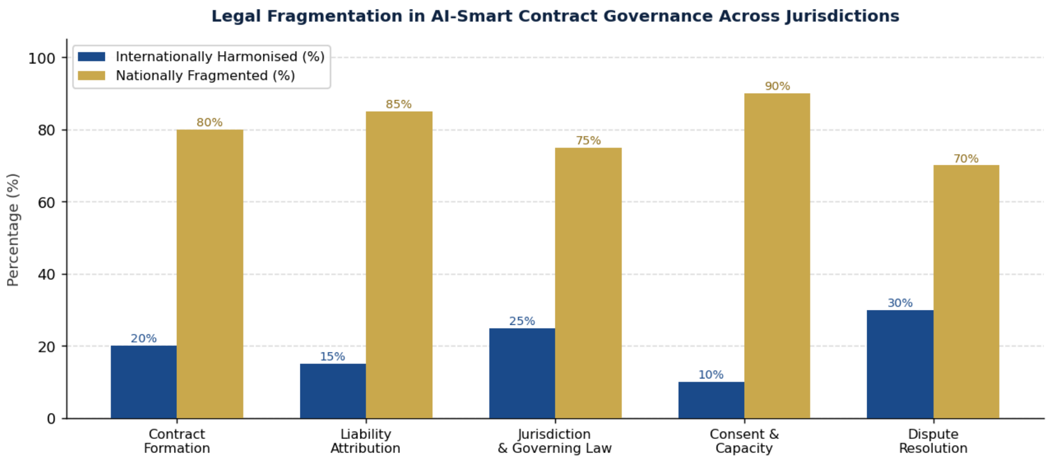

Figure 1: Degree of international harmonisation vs national fragmentation across key AI smart contract governance dimensions (illustrative, based on comparative regulatory mapping).

2. The Legal Gap: Where Existing Frameworks Fall Short

AI-driven smart contracts traverse the entire lifecycle of a commercial transaction — from offer initiation to breach resolution — generating distinct legal questions at each stage that existing instruments leave unresolved.

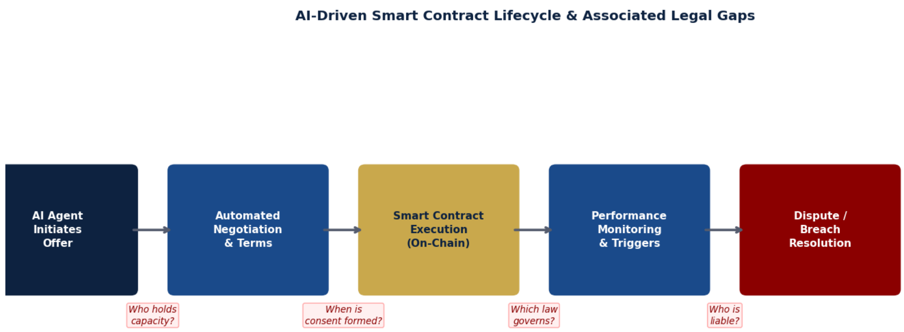

Figure 2: AI-driven smart contract lifecycle and the legal gaps arising at each transition point.

2.1 Legal Capacity of AI Agents

The foundational requirement of legal capacity — the ability to hold rights and incur obligations — is unsettled for AI systems across all major jurisdictions. The PICC Article 1.3 presupposes parties with legal standing, yet AI agents are neither natural nor legal persons under any existing international instrument. Without harmonised rules, courts default to domestic law, producing irreconcilable outcomes in multi-jurisdictional transactions.

2.2 Contract Formation and the Consent Problem

Classical contract formation requires a meeting of minds — offer and acceptance — grounded in the intent of identifiable human parties. Where AI agents autonomously generate contractual terms, the moment and validity of consent become legally ambiguous. Existing UNCITRAL e-commerce instruments address automated message systems but do not contemplate fully autonomous AI negotiation, leaving a critical gap for high-frequency algorithmic contract-making in financial markets.

2.3 Liability Attribution

Perhaps the most commercially significant gap concerns liability: when an AI-executed contract produces harm — through algorithmic error, data manipulation, or unexpected self-execution — who bears responsibility? The EU AI Act (2024) introduces a provider/deployer distinction, but this framework applies only within the EU and does not integrate with international private law instruments. A party in India contracting with a counterpart in Brazil through an AI system hosted in Singapore faces three potentially incompatible liability regimes simultaneously.

2.4 Jurisdiction and Choice of Law

Smart contracts are by design jurisdiction-neutral — they execute on decentralised networks with no fixed seat. This renders traditional connecting factors (domicile, place of contract, place of performance) difficult or impossible to apply. Existing instruments, including the Hague Principles on Choice of Law (2015), do not address the scenario where no human party makes an active choice of law — the AI system simply executes according to its encoded parameters.

2.5 Dispute Resolution

AI-executed contracts often include encoded dispute resolution mechanisms — automatic payment withholding, algorithmic arbitration triggers — whose enforceability under the New York Convention (1958) is untested. Parties from developing economies, who may lack sophisticated legal infrastructure, are particularly vulnerable to opaque, unappealable AI-driven outcomes.

3. Comparative Analysis of Existing Instruments

A review of the primary existing frameworks reveals that while each addresses elements of digital commerce, none provides a comprehensive, internationally harmonised solution for AI-driven smart contracts.

Comparative Analysis: Existing Frameworks vs Proposed UNIDROIT Protocol

Framework	Scope	AI Contracts Covered?	Liability Rules	Jurisdiction Clarity	Binding Effect
UNIDROIT PICC (current)	Intl commercial contracts	✗ No	Traditional parties	Limited	Soft law
EU AI Act (2024)	AI system regulation	Partial	Provider/deployer	EU only	Hard law (EU)
UNCITRAL e-Commerce	Electronic transactions	Partial	Not specified	Moderate	Model law
Proposed UNIDROIT Protocol	**AI-executed contracts**	**✓ Yes**	**Attributed by role**	**Clear**	**Model law + Opt-in**

Figure 3: Comparative analysis of existing international frameworks against the proposed UNIDROIT Protocol.

The UNIDROIT PICC, while a landmark achievement in harmonising international commercial contract law, operates on the assumption of human contractual parties. The EU AI Act introduces important risk-based regulation but is geographically limited and oriented toward regulatory compliance rather than private law rights. UNCITRAL's work on electronic commerce provides valuable foundations but predates autonomous AI systems. No existing instrument combines international scope, private law focus, and AI-specific rules — precisely the combination a new UNIDROIT Protocol could provide.

4. Proposed UNIDROIT Model Protocol: A Five-Pillar Framework

This essay proposes that UNIDROIT develop a Model Protocol on AI-Executed International Commercial Contracts, structured around five core pillars that directly address the identified legal gaps:

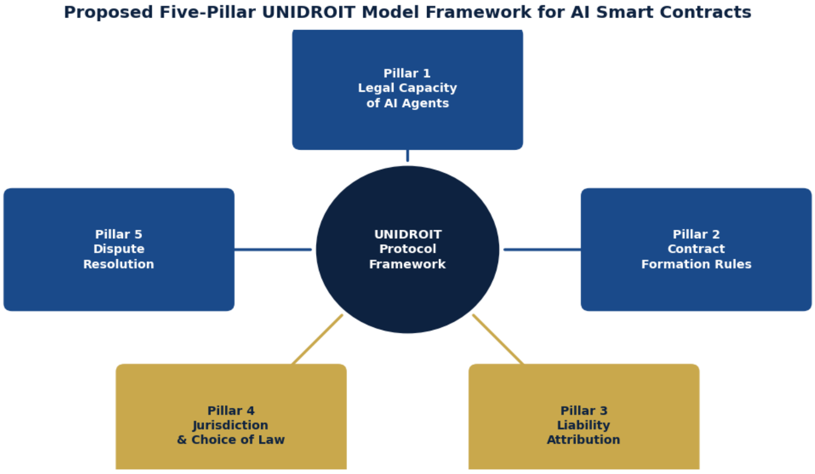

Figure 4: The proposed Five-Pillar UNIDROIT Model Protocol for AI-driven smart contracts in international commerce.

Pillar 1 — Legal Capacity

The Protocol should establish a functional definition of 'AI agent' for private law purposes, attributing limited transactional capacity to AI systems acting within defined parameters, while assigning ultimate legal personhood — and accountability — to the human or corporate principal who deploys the system.

Pillar 2 — Contract Formation

Building on PICC Articles 2.1.1–2.1.11, the Protocol should introduce adapted formation rules recognising AI-generated offers and acceptances as legally valid when: (a) the deploying principal has expressly authorised AI contracting within defined parameters; and (b) the counterparty is notified that it is contracting with an AI system.

Pillar 3 — Liability Attribution

The Protocol should establish a tiered liability chain: primary liability rests with the AI deployer; secondary liability attaches to the developer where harm results from a system defect; and contributory liability may apply to the contracting counterparty where it deliberately exploited AI system limitations.

Pillar 4 — Jurisdiction and Choice of Law

The Protocol should introduce a 'closest connection' test adapted for AI contracts, with a fallback presumption in favour of the law of the AI deployer's place of business, subject to mandatory consumer protection overrides. Parties should be encouraged to include explicit governing law and jurisdiction clauses in AI-enabling agreements.

Pillar 5 — Dispute Resolution

The Protocol should validate AI-encoded dispute resolution mechanisms subject to minimum standards of transparency, human review rights, and accessibility — drawing on UNCITRAL Model Law on International Commercial Arbitration and ensuring compatibility with the New York Convention.

5. Implementation Pathway and Broader Considerations

A Model Protocol — rather than a binding convention — is the appropriate instrument for this stage of legal development, for three reasons. First, it allows jurisdictions to adopt the framework at their own pace, building an international consensus before binding commitments are sought. Second, it enables coordination with ongoing work at UNCITRAL, the European Commission, and national AI regulatory bodies. Third, it follows the successful precedent of UNIDROIT's own Legislative Guide on Secured Transactions and the Cape Town Convention, which used soft-law precursors to build the foundation for hard-law instruments.

Particular attention must be paid to equitable access. Parties from developing economies — including emerging markets in South Asia, Africa, and Latin America — are increasingly participants in cross-border AI-mediated commerce but have the least capacity to navigate fragmented legal regimes. A harmonised UNIDROIT framework would disproportionately benefit these parties by providing predictable, neutral rules they can rely upon regardless of the financial sophistication of their counterparts.

6. Conclusion

The rise of AI-driven smart contracts is not a distant technological possibility — it is a present commercial reality creating daily legal uncertainty for parties across the globe. As UNIDROIT celebrates its centenary, the Institute stands uniquely positioned, by mandate, expertise, and institutional legitimacy, to fill this critical gap in the architecture of international private law.

A targeted Model Protocol addressing legal capacity, contract formation, liability attribution, jurisdiction, and dispute resolution would extend UNIDROIT's century-long mission into the frontier of AI commerce — harmonising private law not merely for the world of today, but for the world of the next hundred years.

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